

ECON 301—ECONOMETRICS I

Spring 2023

Department of Economics
Nazarbayev University

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Lecture time and classroom Tu-Th 16:30–17:45, 9.105

Office hours: By appointment (Tuesdays and Thursdays), book at <https://alejandromelo.youcanbook.me>, please sign at least one day in advance.

Course Description The purpose of this course is to help students understand how to interpret economic data. To that end, you will learn how to do basic empirical research. To do empirical research means using statistical techniques to manage and analyze data. The main goal is to quantify how one economic variable (such as someone's education) affects another variable (for instance his wage). These techniques are used by both academic economists (professors) and professional economists (i.e. those working for business, government and NGOs). Academic economists use them, in large part, to test theories and models; professional economists do empirical work to help management with decision making. You will gain practical experience by solving the problems sets where you will be analyzing actual economic data.

Course Requirements ECON 211—Economic Statistics, OR MATH 321—Probability and either MATH 322—Mathematical Statistics or MATH 310—Applied Statistical Methods.

If you have a working knowledge of the following terms, you are on good foot:

- Population, Sample, Data
- Random Variable, Probability Distribution
- Correlation, Independence
- Moments of a Distribution: Expected Value, Variance, Covariance, Correlation
- Point Estimator, Interval Estimator (Confidence Interval)
- Sampling Distribution of an Estimator
- Properties of Estimators: Unbiasedness, Consistency, Efficiency

It is your responsibility to know the prerequisites! If you need to review this material, please review your notes of ECON 211/MATH 321-322 or the appendix of Wooldridge (see below for the reference).

Course Website All course material will be posted on the Moodle platform, so it is important that you check regularly. I will also create a Telegram group for the class, so that I can communicate quickly and efficiently with you. I will post the link to the Telegram group on the first day of classes.

Course Material There are two textbooks for this class:

1. *Introductory Econometrics: A Modern Approach* by Jeffrey Wooldridge, Cengage Learning, 2020, 7th Edition. I asked the library to provide the latest edition for you in e-book format. However, any edition will suffice. There are sufficient copies of older editions in the library. If you prefer to have a physical copy of the book, please arrange with the library to have it delivered to you.

2. *A First Course in Econometrics* by Arthur Goldberger, Harvard University Press, 1991. This is a classic textbook in Econometric Theory. However, it is difficult to obtain and the library couldn't obtain enough copies. There are a few copies on the library which I have put on reserve. However, I may be able to provide you with some chapters in pdf when needed.

My course is a combination of some of the development of the topic in Goldberger and Wooldridge. I do not follow either of them to the letter but I will let you know which chapters I am referencing in each lecture.

All other course materials and lecture notes will be posted in Moodle, in particular, I always upload my handwritten notes from the video lecture to the platform, so be sure to check them.

Computational Requirements R with RStudio are the recommended software packages. They are free to use. You are expected to familiarize yourself with this package without losing your focus on the main material in the course. If there is another software package you are already familiar with or more interested in learning (EViews, STATA, SAS, MATLAB, etc.), you might use that package for your homework assignments. Assignments as well as exams will include R output.

Course Grade I will grade the course based on the following items:

Components	Weight
Participation	5%
Problem Sets	10%
Midterm	30%
Final Exam	35%
Research Project (more on this in the next section)	20%

Participation will be assessed via pop-up quizzes and participation during class.

In this class there are two type of assignments:

- **Homework:** This item refers to problems or questions that I will ask during the lectures that you must solve on your own. It consists of extensions of the material or filling the gaps in any of the arguments studied during the lecture. This component is ungraded and only for your own benefit, you do not need to submit it. However, anything that I ask as part of the homework can be assessed in the exams.
- **Problem Sets:** There will be five problem sets. You need to submit these. Late submissions will **not** be accepted. I will grade your problem sets on a scale from 0 to 2 where:
 - 0: Not delivered.
 - 1: Delivered but incomplete.
 - 2: Delivered and complete.

I will post solutions to the problem sets immediately after the deadline. *It is your responsibility to check your solutions against my solutions to see if you were correct or where you made mistakes.* Problem sets are difficult and are meant to represent the level of difficulty that you will face in the exam problems.

The midterm exam will take place **between February 19–March 24**. I will discuss with you exactly what day works for everyone. The final exam will be scheduled by the Registrar.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above. The numeric grade will then be curved to obtain the final letter grade.

Students are expected to take all exams at the designated time. Make-up exams and homeworks will **not** be given. If you miss an exam or problem set due to serious illness, or injury of yourself or a close family member, its weight will be shifted to the remaining exams (problem sets) equally.

Research Project Finding a topic is an important part of the exercise: I won't offer suggestions (although I will be happy to respond to preliminary ideas that you come up with). The paper should be short: 5–10 pages (excluding title and reference pages) discussing the econometric analysis of a group of economic or socioeconomic variables of your choice. It should be well written. You are encouraged to use the Writing Center for help with this.

You are encouraged to work together for your research project **with a limit of maximum 5 people** per project. The project is due by **Saturday, April 30, at 5 PM**. The research project grade is determined using the following rubric:

Components	Weight
Introduction and motives for the research	15%
Integration of the literature review into the paper	15%
Model and pretests	20%
Results and inference	20%
Conclusions	15%
Grammar, writing quality	15%

Tentative Course Outline

1. The purpose of Econometrics.
2. Functional Relationships in Econometrics.
3. Simple Linear Regression and the Conditional Approach.
4. Some Large Sample Theory.
5. Estimation of the Multiple Linear Regression Model.
6. Inference in the Multiple Linear Regression Model.
7. Applications.
8. Some Large Sample Theory for Time Series Data (time permitting). *Update: removed since the semester started two weeks later.*
9. Multiple Linear Regression Models with Time Series Data (time permitting). *Update: removed since the semester started two weeks later.*

Rules and Regulations During lectures, please keep your laptops, tablets and cell phones switched off and refrain from text messaging. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviors.

Academic Misconduct Plagiarism or cheating will not be tolerated and sanctioned according to the procedures in the Student Code of Conduct for Nazarbayev University. Students should familiarize themselves with this code. Please note that I do not discuss plagiarism cases unless a student makes an appeal to the disciplinary committee where my original penalty might be decreased or increased.

You can find the university's official statement on plagiarism below:

Plagiarism is intentionally or carelessly presenting the work of another as one's own. It includes submitting an assignment purporting to be the student's original work which has wholly or in part been created by another person. It also includes the presentation of the work, ideas, representations, or words of another person without customary and proper acknowledgement of sources. Plagiarism occurs when a person:

1. *Directly copies one or more sentences of another person's written work without proper citation. If another writer's words are used, you must place quotation marks around the quoted material and include a footnote or other indication of the source of the quotation. This includes cut and paste from the Internet or other electronic sources;*
2. *Changes words but copies the sentence structure of a source without giving credit to the original source, or closely paraphrases one or more paragraphs without acknowledgment of the source of the ideas, or uses graphs, figures, drawings, charts or other visual/audio materials without acknowledging the source or the permission of the author;*
3. *Submits false or altered information in any academic exercise. This may include making up data for an experiment, altering data, citing nonexistent articles, contriving sources, etc.;*
4. *Turns in all or part of assignment done by another student and claims it as their own;*
5. *Uses a paper writing service, has another student write a paper, or uses a foreign language translation and submits it as their own original work.*